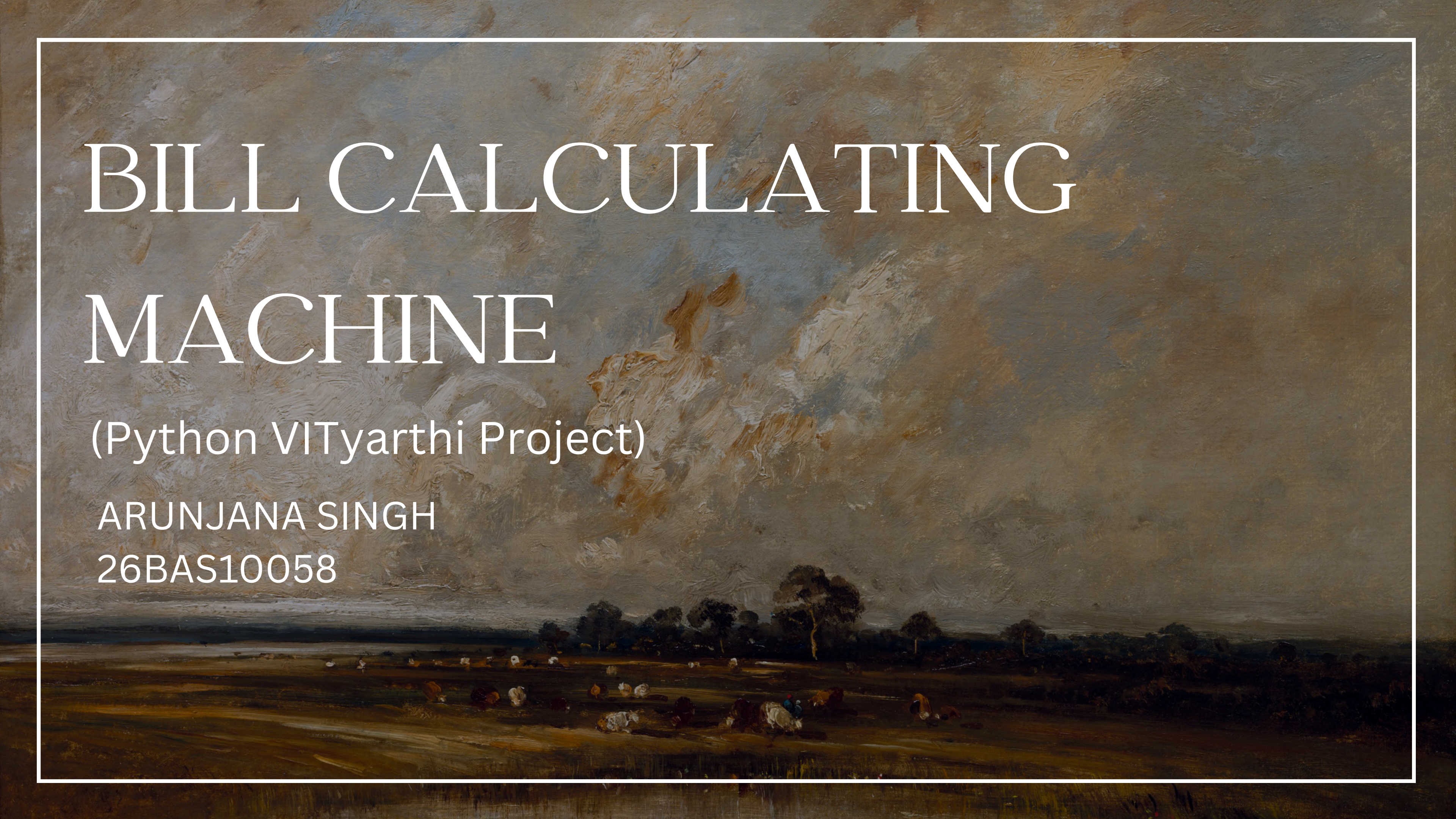


BILL CALCULATING MACHINE

(Python VITyarthi Project)

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INTRODUCTION

This project is a comprehensive desktop-based shop billing and inventory management application built using Python and Tkinter. It provides retail shopkeepers with an intuitive graphical user interface (GUI) to handle point-of-sale checkout operations, track stock in real time, and manage promotional offers. By combining automated subtotal discounts and membership loyalty validation, the application streamlines store operations and minimizes human error in financial calculations.

PROBLEM STATEMENT

Small and medium-sized retail shop owners often struggle with manual billing, inventory tracking inaccuracies, and complex discount applications. Manual billing slows down customer checkouts, increases human error during calculations, and makes tracking real-time stock levels difficult. Existing commercial point-of-sale (POS) systems can be overly expensive or complex for smaller businesses. This project addresses the need for a lightweight, desktop-based billing and inventory management tool that automates transaction calculations, streamlines stock tracking, and simplifies loyalty discount rules



SCOPE OF THE PROJECT

IN SCOPE

- Automated bill generation based on item codes and stock quantities.
- Tiered subtotal discount logic and regex validation for membership discounts.
- Admin panel for managing inventory (adding new products and restocking).
- Real-time persistent stock updates saved to JSON format

OUT OF SCOPE

- Multi-store cloud synchronization.
- Integrated digital payment gateways (credit cards/UPI).

FUNCTIONAL REQUIREMENTS

- FR-1: Navigation Module: Provides a simple GUI for switching between billing, administration, and exit options.
- FR-2: Billing & Cart Module: Allows cashiers to enter item codes and quantities, checks available stock, and calculates line totals dynamically.
- FR-3: Discount & Membership Module: Automatically calculates subtotal tiered discounts (5% for $\geq ₹100$, 10% for $\geq ₹200$, 20% for $\geq ₹500$) and applies an extra 10% discount for validated membership credentials.
- FR-4: Administrative Control Module: Authenticates admin credentials to view inventory records, adjust stock levels, or add new items.

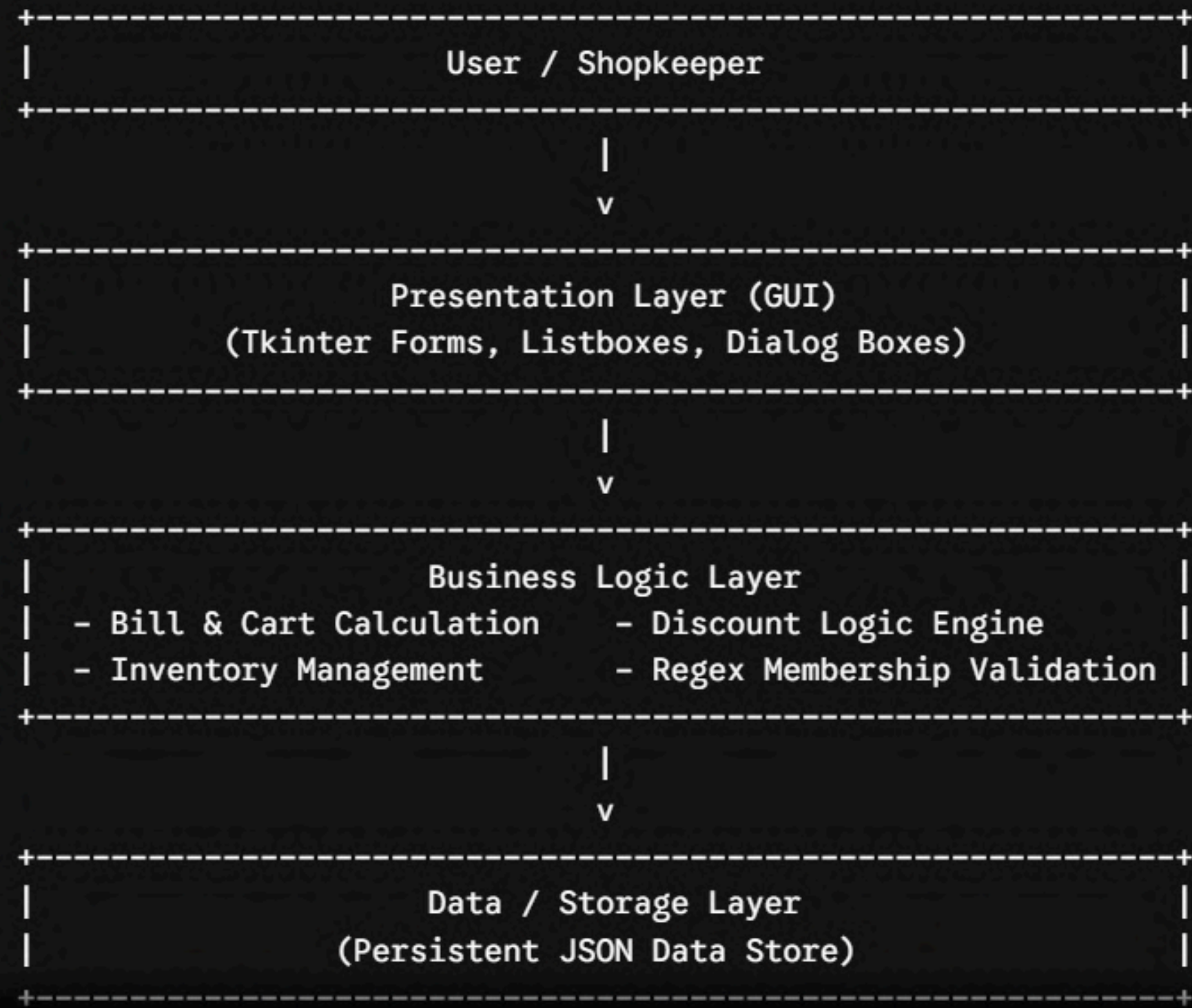


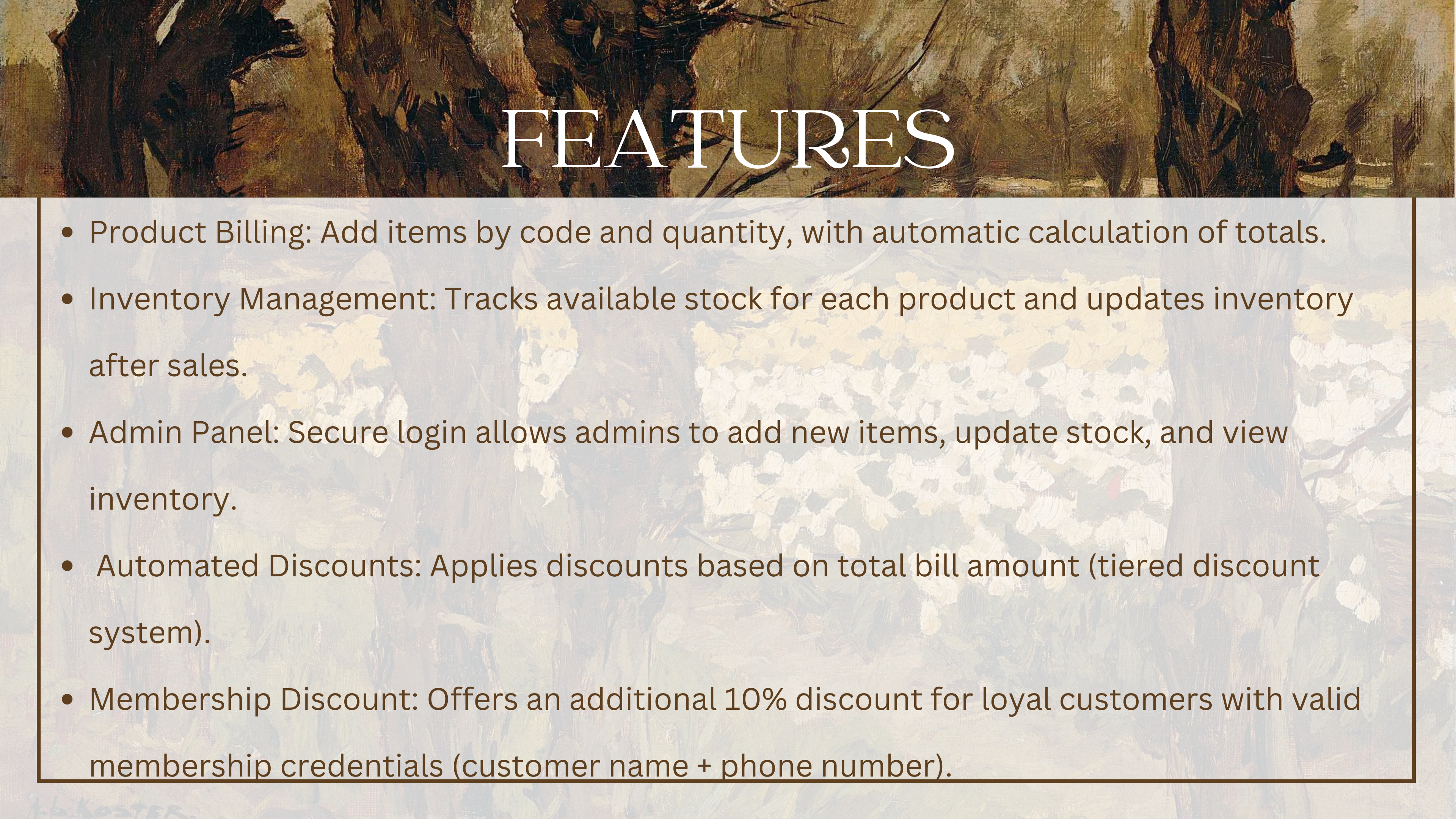
NON FUNCTIONAL REQUIREMENTS

- NFR-1: Performance: Response time for input processing, bill calculation, and stock validation takes less than 1 second.
- NFR-2: Usability: Features an intuitive button-driven navigation interface requiring minimal user training.
- NFR-3: Reliability: Validates user input against non-numeric quantities, non-existent item codes, out-of-stock items, and unauthorized admin logins without application crashes.
- NFR-4: Maintainability: Code base is modularized across 6 distinct files across logical packages for ease of maintenance. .



SYSTEM ARCHITECTURE

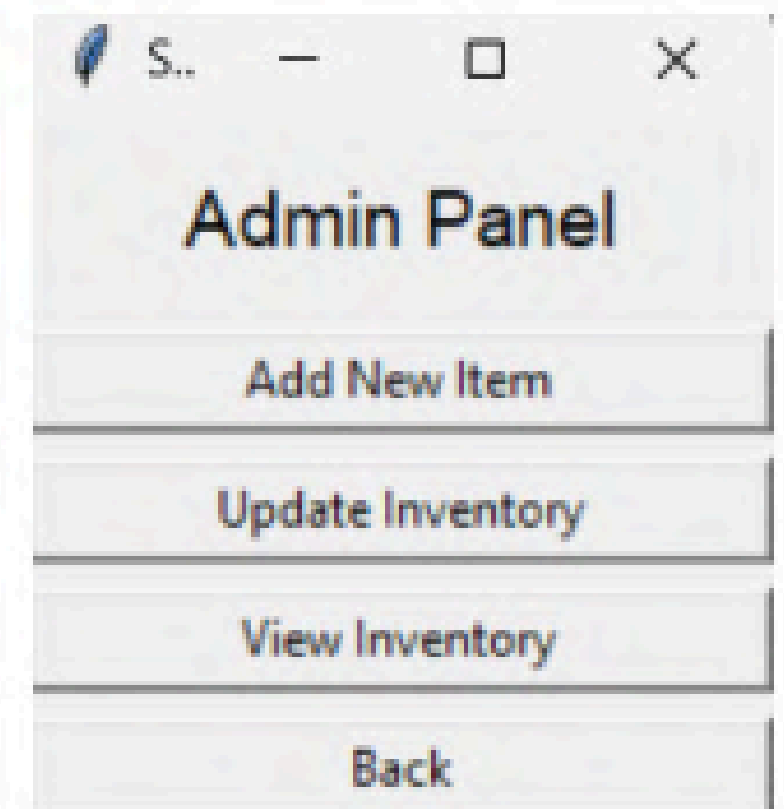
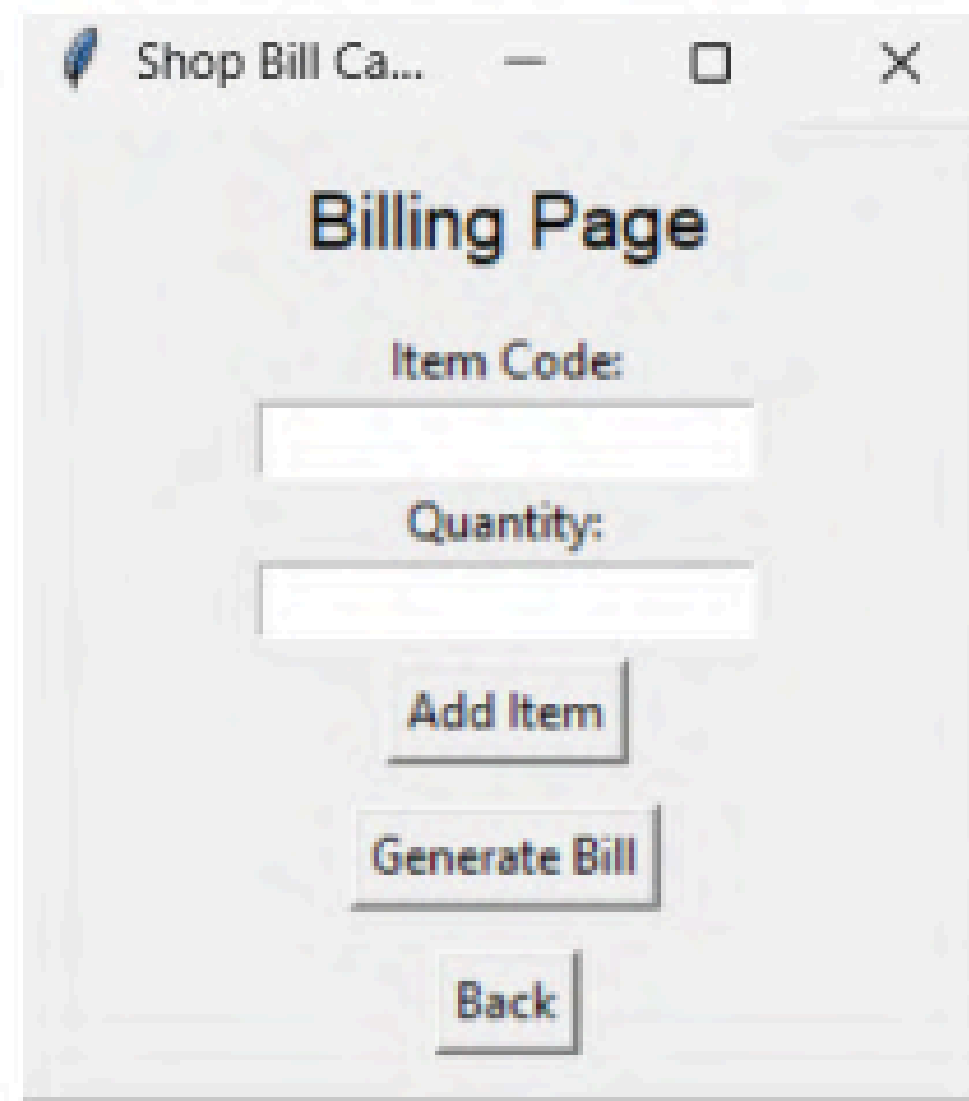
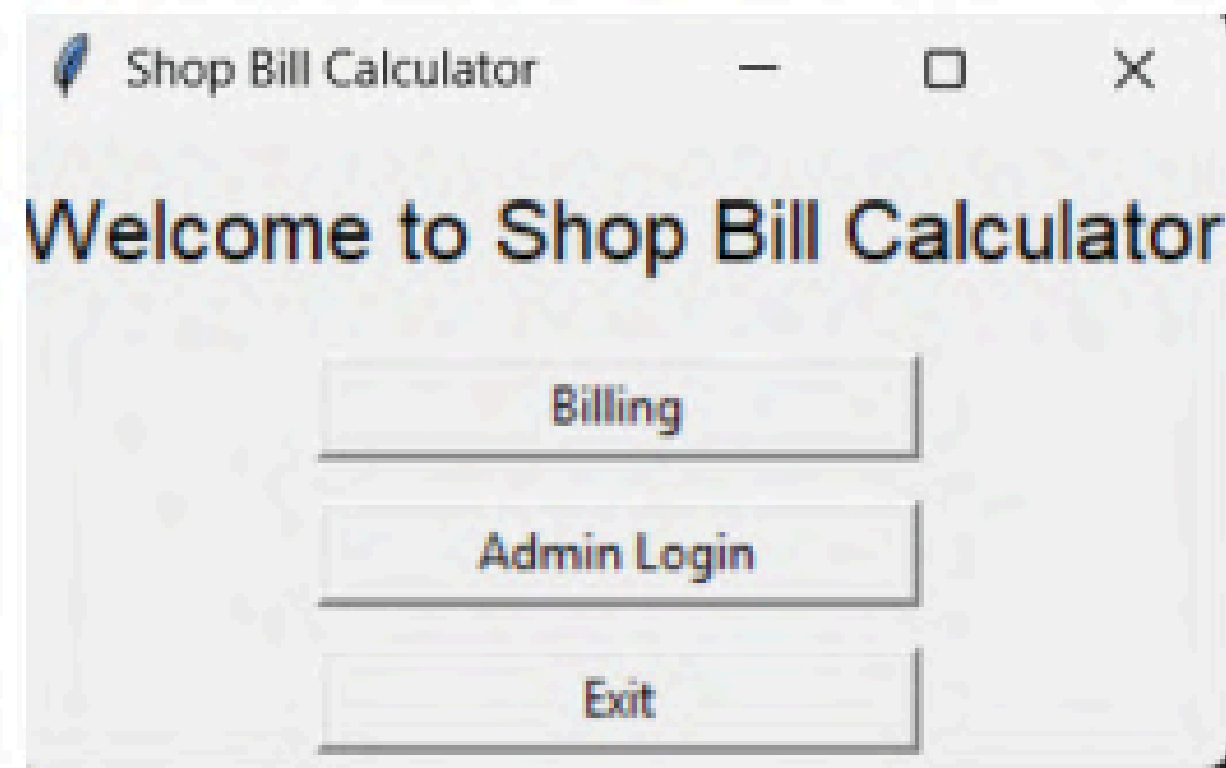




FEATURES

- Product Billing: Add items by code and quantity, with automatic calculation of totals.
- Inventory Management: Tracks available stock for each product and updates inventory after sales.
- Admin Panel: Secure login allows admins to add new items, update stock, and view inventory.
- Automated Discounts: Applies discounts based on total bill amount (tiered discount system).
- Membership Discount: Offers an additional 10% discount for loyal customers with valid membership credentials (customer name + phone number).

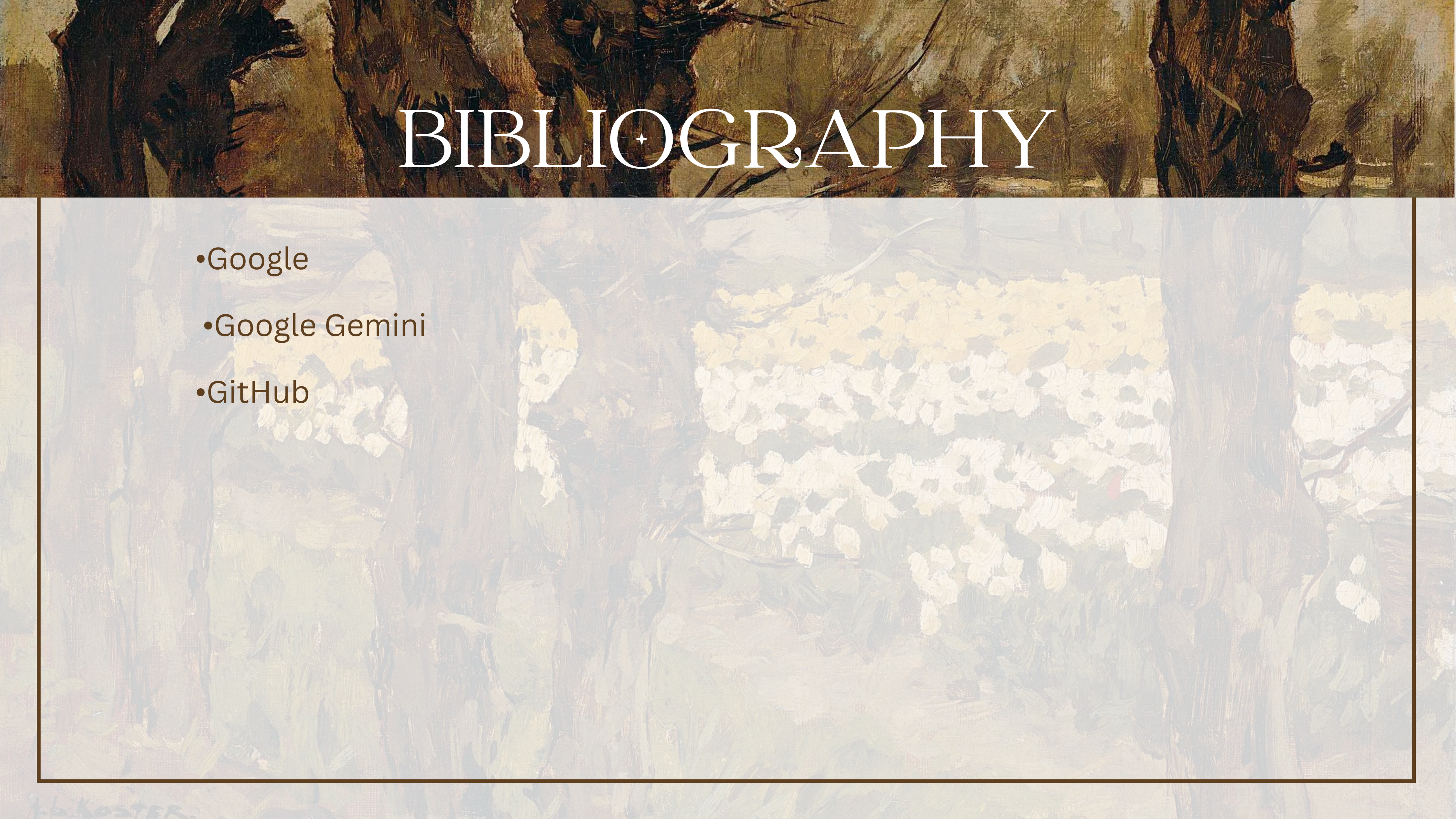
OUTPUTS





CONCLUSION

This project delivers a practical and user-friendly shop billing system with inventory tracking, admin management, and customer rewards. It streamlines billing operations for retail shops, making them more efficient and customer-centric, while providing a solid foundation for future enhancements and scalability

The background of the slide is an oil painting of a forest scene. In the foreground, there are dark, textured tree trunks. A path leads from the bottom center towards the middle ground, where a small boat is visible on a body of water. The sky is a mix of blue and white, suggesting a bright, overcast day. The overall style is impressionistic, with visible brushstrokes and a rich color palette.

BIBLIOGRAPHY

- Google
- Google Gemini
- GitHub

THANK YOU

